

Customer Transactions

Banking Dashboard

2015–2019 | Excel + MySQL + Power BI

■238.37bn Deposits · ■240.20bn Withdrawals · 116,162 Records · 10 Accounts · 3 Tools

TOOLS	SOURCE	ACCOUNTS	DATE RANGE	RECORDS
Excel+MySQL+Power BI	Kaggle (Public)	10 Bank Accounts	Jan 2015–Mar 2019	116,162 · 14 Cols

01 EXECUTIVE SUMMARY

- Processed 116,201 raw banking transaction records through a 3-tool pipeline (Excel → MySQL → Power BI) — removing 39 exact duplicates in Excel, standardising date formats to YYYY-MM-DD, engineering 5 derived columns (CATEGORY, TXN_TYPE, AMOUNT, TXN_YEAR, TXN_MONTH), and importing 116,162 clean records into MySQL via LOAD DATA LOCAL INFILE after resolving 4 critical import errors.
- Designed a MySQL database (banking_project) with 1 transaction table (14 columns) and 1 analytical VIEW (vw_monthly_summary), executed 7 analytical queries covering KPIs, monthly trends, category breakdown, account performance, year-over-year comparison, and deposit vs withdrawal splits — creating a complete SQL-based analytical layer before connecting to Power BI.
- Built a 3-page interactive Power BI dashboard (18 visuals total) with 7 DAX measures, Sync Slicers across pages, and a custom Banking Premium Navy & Gold theme — covering Overview (Page 1), Category & Accounts (Page 2), and Trends & Flow (Page 3) — all connected live to MySQL.
- Analyzed ■238.37bn total deposits and ■240.20bn total withdrawals across 116,162 transactions (10 accounts, Jan 2015–Mar 2019) — surfacing ■-1.83bn net negative cash flow, 2016 as peak year (■99.72bn deposits, 30,367 transactions), Fund Transfer as dominant category (28.50% of transactions, 70.10% of total amount), and 97.5% of all balance records in negative territory indicating persistent account overdraft.

02 PROBLEM STATEMENT

Banking transaction data for 10 accounts spanning January 2015 to March 2019 existed as a raw Excel file with 116,201 records containing no structured reporting layer, no transaction categorisation, no derived analytical columns, and several data quality issues — 39 duplicate rows, currency-formatted amounts stored inconsistently, and dates in non-MySQL-compatible formats. Management had no visibility into category-wise spending patterns, account-level performance comparisons, monthly flow trends, or the growing negative balance position across a ■478.57bn total transaction portfolio. This project bridges raw transactional data to actionable banking insights using a structured 3-tool intermediate pipeline.

03 OBJECTIVES

- Obj 1: Clean and standardise raw banking data in Excel — remove duplicates, fix date formats, engineer 5 analytical columns (CATEGORY using CHQ.NO. + keyword detection, TXN_TYPE, AMOUNT, TXN_YEAR, TXN_MONTH).

- Obj 2: Build a MySQL relational database — create banking_project database, bank_transactions table with 14 typed columns, import 116,162 rows via LOAD DATA LOCAL INFILE, and execute 7 analytical queries.
- Obj 3: Create 1 analytical VIEW (vw_monthly_summary) enabling pre-aggregated monthly analysis for Power BI performance.
- Obj 4: Connect Power BI to MySQL, apply correct data types in Power Query, create 7 DAX measures including Cheque Transactions count.
- Obj 5: Build a professional 3-page dashboard with 18 visuals covering KPIs, trends, category breakdown, account analysis, and balance trajectory with Sync Slicers across all pages.
- Obj 6: Surface key banking insights — net cash flow position, peak transaction periods, category concentration, cheque vs digital payment split, and account-level overdraft risk.

04 DATASET OVERVIEW

Attribute	Detail
Dataset Name	Bank Transaction Data
Source	Kaggle — apoorwatsky/bank-transaction-data
File Format	Microsoft Excel Workbook (.xlsx)
Raw Records	116,201 rows · 9 original columns
Clean Records	116,162 rows (39 exact duplicates removed)
Date Coverage	January 1, 2015 – March 5, 2019 (4 years, 2 months)
Bank Accounts	10 unique accounts
Total Derived Columns	5 added in Excel: CATEGORY, TXN_TYPE, AMOUNT, TXN_YEAR, TXN_MONTH
Final MySQL Columns	14 (9 original + 5 derived, including CHQ.NO. retained)
Key Data Issue	Date format 'DD-Mon-YY' incompatible with MySQL → reformatted to YYYY-MM-DD
Second Issue	Empty cells in WITHDRAWAL AMT / DEPOSIT AMT → MySQL strict mode rejection
Resolution	LOAD DATA LOCAL INFILE after enabling OPT_LOCAL_INFILE=1

Column Definitions (14 Final Columns in MySQL)

Column	Type	Source	Description
account_no	VARCHAR(30)	Original	Bank account identifier (10 unique accounts)
txn_date	DATE	Original	Transaction date (YYYY-MM-DD format)
txn_details	VARCHAR(255)	Original	Transaction description — used for CATEGORY keyword detection
chq_no	VARCHAR(30)	Original (KEPT)	Cheque number — NULL = digital, value = cheque payment
value_date	DATE	Original	Value/settlement date

Column	Type	Source	Description
withdrawal_amt	DECIMAL(18,2)	Original	Withdrawal amount (NULL when row is deposit)
deposit_amt	DECIMAL(18,2)	Original	Deposit amount (NULL when row is withdrawal)
balance_amt	DECIMAL(18,2)	Original	Account balance after transaction (97.5% negative)
category	VARCHAR(50)	Derived-Excel	Transaction category (11 categories using CHQ.NO. + keywords)
txn_type	VARCHAR(15)	Derived-Excel	DEPOSIT or WITHDRAWAL based on which AMT column has value
amount	DECIMAL(18,2)	Derived-Excel	Unified amount: takes WITHDRAWAL AMT or DEPOSIT AMT value
txn_year	INT	Derived-Excel	Year extracted from txn_date (2015–2019)
txn_month	VARCHAR(15)	Derived-Excel	Month-Year string (e.g. Jun-2017) for display

05 TOOLS & TECHNOLOGIES USED

Tool	Role in Project	Key Actions
Microsoft Excel	PHASE 1 — Data Cleaning & Engineering	Duplicate removal (39 rows), date format fix, 5 derived columns, CSV export
MySQL 8.0	PHASE 2 — Database & SQL Analysis	CREATE DATABASE, CREATE TABLE, LOAD DATA INFILE, 7 queries, 1 VIEW
MySQL Workbench	MySQL GUI Client	Schema design, import wizard, query execution, result verification
Power BI Desktop	PHASE 3 — Dashboard & Visualisation	MySQL connection, Power Query type fixes, 7 DAX measures, 3-page dashboard
DAX	Power BI Calculation Language	7 measures: deposits, withdrawals, net flow, transactions, balance, cheque count
Sync Slicers	Power BI Feature	Year/TXN Type/Category slicers on Page 1 synced to all 3 pages
JSON Theme	Power BI Custom Theme	Banking Premium Navy & Gold — dark navy canvas, gold accents

06 PHASE 1 — EXCEL: DATA CLEANING & ENGINEERING

Step	Action in Excel	Detail / Outcome
01	Open bank.xlsx — explore structure	9 columns: Account No, DATE, TRANSACTION DETAILS, CHQ.NO., VALUE DATE, WITHDRAWAL AMT, DEPOSIT AMT
02	Delete trailing '.' column	Junk separator column removed — 8 usable columns remain
03	Keep CHQ.NO. column (strategic decision)	CHQ.NO. non-null = CHEQUE category; used in CATEGORY formula as primary check

Step	Action in Excel	Detail / Outcome
04	Fix Account No apostrophe	Ctrl+H → find ' → replace with empty; removed BOM apostrophe from all 116,201 account numbers
05	Fix Date format to YYYY-MM-DD	Format Cells → Custom → YYYY-MM-DD applied to DATE and VALUE DATE columns for MySQL compatibility
06	Create CATEGORY column (col I)	IF(D2<>"",\ 'CHEQUE\ ', nested SEARCH() for 14 keywords → 11 categories: FUND TRANSFER, ATM/CASH, UPI PAYMENT
07	Create TXN_TYPE column (col J)	=IF(F2>0,"WITHDRAWAL","DEPOSIT") — identifies transaction direction
08	Create AMOUNT column (col K)	=IF(F2>0,F2,G2) — unified single amount field from either withdrawal or deposit column
09	Create TXN_YEAR column (col L)	=YEAR(B2) — extracted year for time-series analysis (2015–2019)
10	Create TXN_MONTH column (col M)	=TEXT(B2,"MMM-YYYY") — display month string (e.g. Jun-2017)
11	Remove duplicates	Data → Remove Duplicates → ALL columns selected → 39 exact duplicates removed → 116,162 unique rows
12	Save as CSV UTF-8	File → Save As → CSV UTF-8 → bank_transactions_clean.csv

07 PHASE 2 — MYSQL: DATABASE, IMPORT & QUERIES

Database & Table Creation (SQL)

Created banking_project database and bank_transactions table with 14 typed columns including chq_no (VARCHAR) retained from the original dataset. All amount columns defined as VARCHAR initially to handle empty-string import errors from MySQL strict mode, with subsequent ALTER TABLE to correct types.

Import Method — LOAD DATA LOCAL INFILE

- Table Data Import Wizard (GUI): Attempted first — rejected due to date format incompatibility (Error: Incorrect date value '21-Oct-16') and decimal mode rejections for empty cells.
- LOAD DATA LOCAL INFILE (Final method): Required enabling OPT_LOCAL_INFILE=1 in MySQL Workbench Advanced connection settings and SET GLOBAL local_infile=1 — imported all 116,162 rows successfully in under 60 seconds vs 9+ hours estimated by the GUI wizard.

7 Analytical Queries Executed

Query	Purpose	Key Result
Q1 — Overall KPIs	Total deposits, withdrawals, net flow, avg balance, row count	■238.37bn deposits, ■240.20bn withdrawals, - ■1.83bn net flow, 116,162 rows
Q2 — Monthly Trend	Deposits, withdrawals, net flow grouped by year and month	April 2016 highest deposits (■9.92bn); July 2018 peak transaction count (3,969)

Query	Purpose	Key Result
Q3 — Category Breakdown	Count, deposit, withdrawal, total amount per category	Fund Transfer = 28.50% of txns but 70.10% of total amount; OTHER = 52.91% of txns
Q4 — Account Performance	Deposits, withdrawals, min/max/avg balance per account	Account 409000362497 highest deposits (■101.72bn); 1196428 most transactions (48,758)
Q5 — Year-over-Year	Annual deposits, withdrawals, transaction count, avg balance	2016 peak: 30,367 txns, ■99.72bn deposits; 2018 most transactions: 35,517
Q6 — Deposit vs Withdrawal Split	Count and total amount per TXN_TYPE	62,637 deposits (53.92%), 53,525 withdrawals (46.08%)
Q7 — CREATE VIEW	vw_monthly_summary — pre-aggregated monthly data for Power BI	10-column view: txn_year, month_num, month_name, category, txn_type, account_no, txn_count, deposits, withdrawals, avg_ba

08 PHASE 3 — POWER BI: CONNECTION, DAX & DASHBOARD

MySQL Connection

- Connected Power BI to MySQL via Home → Get Data → MySQL Database → Server: localhost → Database: banking_project.
- Imported 2 objects: bank_transactions (main table, 116,162 rows, 14 columns) and vw_monthly_summary (pre-aggregated VIEW, 10 columns).
- Fixed data types in Power Query: txn_year changed from Text (ABC) to Whole Number; txn_date verified as Date; all amount columns confirmed as Decimal.

7 DAX Measures Created

Measure	DAX Formula	Result
Total Deposits	SUM('banking_project bank_transactions'[deposit_amt])	■238,368,715,323.44
Total Withdrawals	SUM('banking_project bank_transactions'[withdrawal_amt])	■240,201,132,284.82
Net Flow	[Total Deposits] - [Total Withdrawals]	-■1,832,416,961.38
Total Transactions	COUNTROWS('banking_project bank_transactions')	116,162
Avg Balance	AVERAGE('banking_project bank_transactions'[balance_amt])	-■1,404,802,148
Avg Transaction Amount	AVERAGE('banking_project bank_transactions'[amount])	■3,805,557 (dep avg)
Cheque Transactions	COUNTROWS(FILTER(..., chq_no <> BLANK()))	905 (0.78% of all txns)

3-Page Dashboard Structure (18 Visuals)

Page	Visual Type	Fields
Overview	Total Deposits KPI Card	Total Deposits measure
Overview	Total Withdrawals KPI Card	Total Withdrawals measure
Overview	Total Transactions KPI Card	Total Transactions measure
Overview	Net Flow KPI Card	Net Flow measure
Overview	Transaction Volume Trend Line Chart	txn_date → Total Transactions
Overview	Yearly Deposits vs Withdrawals Clustered Column	txn_year → Total Deposits + Total Withdrawals
Overview	Slicer: Year	txn_year (synced to all 3 pages)
Overview	Slicer: Transaction Type	txn_type (synced to all 3 pages)
Overview	Slicer: Category	category (synced to all 3 pages)
Category & Accounts	Cheque vs Digital Split Donut	category (Cheque vs Digital legend)
Category & Accounts	Transaction Volume by Account Bar Chart	account_no → Total Transactions
Category & Accounts	Transaction Breakdown by Category Bar Chart	category → Total Transactions
Category & Accounts	Deposits vs Withdrawals Split Donut	txn_type → Total Transactions
Trends & Flow	Yearly Deposits vs Withdrawals Clustered Column	txn_year → Deposits + Withdrawals
Trends & Flow	Monthly Deposits vs Withdrawals Clustered Column	txn_month → Deposits + Withdrawals
Trends & Flow	Average Balance Trend Line Chart	txn_date → Avg Balance
Trends & Flow	Slicer: Account No	account_no (page 3 only)
All Pages	Sync Slicers Feature	Year, TXN Type, Category synced across Pages 1–3

09 KEY INSIGHTS & RESULTS

A. Overall Financial Position

- Analyzed ■238,368,715,323.44 total deposits and ■240,201,132,284.82 total withdrawals across 116,162 transactions spanning 4 years 2 months — recording a net negative cash flow of ■-1,832,416,961.38, with 97.5% of all 116,162 balance records in negative territory (avg balance ■-1.40bn).
- Maximum single deposit: ■544,800,000 | Maximum single withdrawal: ■459,447,546 | Average deposit transaction: ■3,805,557 | Average withdrawal transaction: ■4,487,643 — withdrawals averaging 17.9% higher than deposits per transaction, contributing to the persistent negative net flow.

B. Year-over-Year Analysis

- 2016 was the peak year for transaction VALUE at ₹99.72bn deposits and ₹100.11bn withdrawals across 30,367 transactions; 2018 was peak for transaction VOLUME at 35,517 transactions but only ₹28.54bn deposits — indicating higher frequency of smaller-value transactions in later years.
- 2019 (partial Jan–Mar only): 5,520 transactions with ₹1.08bn deposits — annualised this suggests a continued declining trend from 2018 levels, consistent with the overall portfolio wind-down pattern observed across all accounts.

C. Transaction Category Analysis

- Fund Transfer dominated by VALUE — 28.50% of transactions but 70.10% of total amount (₹297.20bn combined deposits + withdrawals), confirming large institutional-scale inter-bank transfers as the primary financial activity across these accounts.
- OTHER category dominated by COUNT — 52.91% of all transactions (61,457 rows) but only 16.99% of total amount — indicating high volumes of small undescribed transactions; represents the largest data quality opportunity for re-categorisation with additional keyword rules.
- Cheque transactions: only 905 (0.78%) — confirming overwhelming shift to digital payments (115,257 transactions, 99.22%); UPI Payment at only 16 transactions indicates dataset predates widespread UPI adoption in India.

D. Account-Level Analysis

- Account 1196428 was most active by transaction count (48,758 transactions — 41.97% of all records) with ₹68.30bn deposits and ₹68.45bn withdrawals — nearly perfectly balanced but with avg balance of -₹1.67bn, indicating persistent structural overdraft.
- Account 409000362497 handled highest absolute value — ₹101.72bn deposits and ₹101.94bn withdrawals across 29,840 transactions — the highest value account representing a major institutional fund flow node.
- Account 409000405747 was least active (only 51 transactions) but showed the most extreme negative position (-₹476.68bn avg balance) — a significant risk flag requiring investigation as withdrawal amount (₹420.32bn) massively exceeded deposit amount (₹228.79M).

E. KPI Summary

KPI	Value	KPI	Value
Total Deposits	₹238,368,715,323	Total Withdrawals	₹240,201,132,285
Net Cash Flow	-₹1,832,416,961	Total Transactions	116,162
Deposit Transactions	62,637 (53.92%)	Withdrawal Transactions	53,525 (46.08%)
Avg Balance	-₹1,404,802,148	Max Balance	₹8,500,000
Min Balance	-₹2,045,201,142	Neg Balance Rows	113,237 (97.5%)
Max Single Deposit	₹544,800,000	Max Single Withdrawal	₹459,447,546
Peak Year (Value)	2016 — ₹99.72bn deposits	Peak Year (Volume)	2018 — 35,517 transactions
Peak Month (Txns)	July 2018 — 3,969 transactions	Peak Month (Deposits)	April 2016 — ₹9.92bn

KPI	Value	KPI	Value
Top Category (Count)	OTHER — 61,457 (52.91%)	Top Category (Value)	Fund Transfer — ■297.20bn
Most Active Account	1196428 — 48,758 txns	Cheque Transactions	905 (0.78%)
Digital Transactions	115,257 (99.22%)	Unique Accounts	10

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — MySQL Date Format Rejection (Error: Incorrect date value '21-Oct-16')

- Problem: Raw Excel dates in 'DD-Mon-YY' format (e.g. '16-Aug-17') were rejected by MySQL with 'Incorrect date value' error during the first import attempt via the Table Data Import Wizard, causing complete import failure.
- Solution: Reformatted DATE and VALUE DATE columns in Excel using Format Cells → Custom → YYYY-MM-DD. This converted all dates to MySQL's required ISO format (e.g. '2017-08-16') before re-exporting as CSV and reimporting.

Challenge 2 — MySQL Strict Mode Rejecting Empty Cells in Amount Columns

- Problem: WITHDRAWAL AMT and DEPOSIT AMT columns contain empty/blank cells by design (each row is either a deposit OR withdrawal, so the other column is blank). MySQL strict mode rejected these blank cells as 'Incorrect decimal value' errors.
- Solution: Recreated the table with all amount columns defined as VARCHAR(30) instead of DECIMAL — accepted any value including blanks. Subsequently used ALTER TABLE to convert types and verified data integrity. Ultimately resolved by using LOAD DATA LOCAL INFILE which handled blank cells as NULL automatically.

Challenge 3 — LOAD DATA LOCAL INFILE Disabled (Error Code 3948 & 2068)

- Problem: LOAD DATA LOCAL INFILE blocked by two-level restriction — Error 3948 (server-side disabled) and Error 2068 (client-side disabled) — preventing fast bulk import despite SET GLOBAL local_infile=1 fixing the server side.
- Solution: Edited MySQL Workbench connection settings → Advanced tab → Others box → added OPT_LOCAL_INFILE=1. This enabled the client-side permission. Combined with SET GLOBAL local_infile=1 on the server side, the full LOAD DATA LOCAL INFILE command then executed successfully, importing 116,162 rows in under 60 seconds.

Challenge 4 — Import Wizard Taking 9+ Hours (200 rows/minute)

- Problem: The MySQL Table Data Import Wizard was processing only ~200 rows per minute — estimating 9+ hours to complete importing 116,162 rows, making GUI-based import completely impractical.
- Solution: Switched from the GUI Import Wizard to LOAD DATA LOCAL INFILE command — a server-side bulk loader that processes all rows in a single database operation rather than individual INSERT statements. Completed the same import in under 60 seconds — a 500x+ speed improvement.

Challenge 5 — BOM Character in CSV Causing Column Mapping Error

- Problem: Saving as 'CSV UTF-8' in Excel added a Byte Order Mark (BOM) character to the first column header — making it appear as 'ï»¿Account No' in the MySQL import wizard instead of 'Account No', causing all column mappings to shift by one position.
- Solution: Manually remapped 'ï»¿Account No' to account_no in the import wizard column mapping screen. The BOM is an invisible encoding marker harmless to data — it only affected the header label, not the actual data values.

Challenge 6 — DAX Formula Red Line (Table Name with Spaces)

- Problem: Writing DAX measure 'Total Deposits = SUM(bank_transactions[deposit_amt])' showed a red underline error because the Power BI table name 'banking_project bank_transactions' contains spaces — DAX requires space-containing names to be wrapped in single quotes.
- Solution: Added single quotes around the full table name in every DAX formula: SUM('banking_project bank_transactions'[deposit_amt]). Applied this pattern to all 7 measures — single quotes are mandatory whenever a table or column name contains spaces or special characters.

Challenge 7 — 14 Visuals on 1 Page (Crowded Layout)

- Problem: Initial plan placed all 14 visuals on a single page — creating a severely crowded, unreadable layout that would look unprofessional in a portfolio context and make cross-visual comparison difficult.
- Solution: Restructured into a professional 3-page layout using Power BI's multi-page feature: Page 1 (Overview — KPIs + trends + slicers), Page 2 (Category & Accounts — breakdowns), Page 3 (Trends & Flow — time series). Implemented Sync Slicers via View → Sync Slicers so Year/TXN Type/Category filters on Page 1 automatically apply to Pages 2 and 3.

11 SKILLS LEARNED & TAKEAWAYS

- 3-Tool Data Pipeline (Excel → MySQL → Power BI): Built first end-to-end intermediate pipeline where each tool plays a specific role — Excel for cleaning, MySQL for storage and SQL analysis, Power BI for visualisation — understanding how professional data stacks separate concerns.
- Transaction Categorisation with CHQ.NO.: Used physical cheque number column as the primary category signal (non-null = CHEQUE) before keyword scanning — a real-world banking data pattern where structured fields take precedence over text parsing.
- MySQL Bulk Import — LOAD DATA LOCAL INFILE: Mastered the fastest MySQL import method; understood the two-level permission system (server + client) and why GUI import wizards are unsuitable for large datasets.
- MySQL VIEW Creation: Created vw_monthly_summary as a pre-aggregated analytical view — understanding how VIEWS can simplify Power BI queries, improve performance, and provide a stable reporting interface over complex underlying tables.
- DAX with Space-Containing Table Names: Learned that Power BI table names inheriting MySQL schema+table naming (e.g. 'banking_project bank_transactions') require single-quote wrapping in every DAX formula — a critical syntax rule not encountered in single-table beginners projects.
- Multi-Page Dashboard with Sync Slicers: Designed a 3-page professional dashboard layout and implemented Sync Slicers — understanding the difference between Sync (slicer affects the page) and Visible (slicer appears on the page) settings for cross-page filter architecture.

- Banking Data Interpretation: Learned to read banking signals — net negative cash flow indicates spending excess over income; 97.5% negative balance rows indicate systemic overdraft; Fund Transfer dominating total amount vs OTHER dominating count tells two different stories about the same portfolio.

12 CONCLUSION

- Processed 116,201 raw banking records through a complete 3-tool intermediate pipeline — cleaned 39 duplicates and engineered 5 derived columns in Excel, imported 116,162 rows into MySQL via LOAD DATA LOCAL INFILE after resolving 4 critical import errors, executed 7 analytical SQL queries and created 1 aggregated VIEW.
- Engineered 7 DAX measures including a Cheque Transactions counter and Net Flow calculation, built a 3-page Power BI dashboard with 18 visuals and Sync Slicers connecting all pages — connected live to MySQL rather than importing static data, demonstrating a production-grade BI architecture.
- Surfaced ■-1.83bn net negative cash flow, 2016 as peak value year (■99.72bn deposits), 2018 as peak volume year (35,517 transactions), Fund Transfer dominating 70.10% of total amount on only 28.50% of transaction count, and account 409000405747 as the highest-risk account with ■-476.68bn average balance.
- Identified 97.5% of all balance records as negative, 99.22% of all transactions as digital (only 905 cheques), April 2016 as the highest single-month deposit period (■9.92bn), and July 2018 as peak transaction month (3,969) — all delivered through a professional 3-page banking dashboard with gold-navy premium theme.
- First Intermediate project of the portfolio — demonstrated ability to architect and execute a multi-tool data pipeline, write production SQL queries, design a multi-page BI dashboard with cross-page sync, and extract meaningful financial insights from raw transactional banking data. Project 4 of 9.